

Examen Practico 1

1. Security Group EC2

Outbound security group rules successfully modified on security group (sg-0fe9713b514909fba | EC2-valentina)

sg-0fe9713b514909fba - EC2-valentina

Details

Security group name EC2-valentina	Security group ID sg-0fe9713b514909fba	Description Security Group EC2	VPC ID vpc-00f479057476a2d08
Owner 654654478122	Inbound rules count 1 Permission entry	Outbound rules count 0 Permission entries	

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0d0255242e19afed0	IPv4	SSH	TCP	22

2. Security Group RDS

The screenshot displays the AWS Management Console interface for a Security Group. The left-hand navigation pane shows the 'EC2' section expanded, with 'Security Groups' selected. The main content area shows the details for the Security Group 'sg-05a7bb0abb6959bcf - RDS-valentina'.

Details:

- Security group name:** RDS-valentina
- Security group ID:** sg-05a7bb0abb6959bcf
- Description:** security group RDS
- VPC ID:** vpc-00f479057476a2db8
- Owner:** 654654478122
- Inbound rules count:** 1 Permission entry
- Outbound rules count:** 0 Permission entries

Inbound rules (1):

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-09780236d24d337b0	-	MYSQL/Aurora	TCP	3306

3. Instancia EC2

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 6546-5447-8122

valentina0519@gmail.com

EC2

>

Instances

>

i-0c503328518d7f0cf

Instance summary for i-0c503328518d7f0cf (ec2-valentina-mod2)

Info

Refresh

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-0c503328518d7f0cf

IPv6 address

-

Hostname type

IP name: ip-172-31-4-107.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.220.17.78 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

13.220.17.78 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-4-107.ec2.internal

Instance type

t3.micro

VPC ID

vpc-00f479057476a2db8 (VPC-Produccion)

Subnet ID

subnet-00227b47a5f26ec04

Instance ARN

arn:aws:ec2:us-east-1:654654478122:instance/i-0c503328518d7f0cf

Private IPv4 addresses

172.31.4.107

Public DNS

ec2-13-220-17-78.compute-1.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

User: arn:aws:iam::654654478122:user/students/valentina0519@gmail.com is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action

Retry

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Root device details

Root device name

/dev/xvda

Root device type

EBS

EBS optimization

enabled

Block devices

Filter block devices

Volume ID

Device name

Volume size (GiB)

Volume State

Attachment status

☒

vol-0df633941b17a00e4

/dev/xvda

8

In-use

Attached

☐

vol-06542193b3a431fab

/dev/sdb

3

In-use

Attached

CloudShell

Feedback

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4. RDS

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 6546-5447-8122

valentina0519@gmail.com

Aurora and RDS

Databases

db-valentina

Successfully modified db-valentina.

db-valentina

Modify

Actions

Summary

DB identifier

db-valentina

CPU

4.77%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

us-east-1a

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance

Connect using

Info

☒ Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ Endpoints

Use when connecting through any IDE interface.

Internet access gateway

Disabled

IAM Authentication

Disabled

Programming language

MySQL (macOS)

Endpoint type

Instance endpoint

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

1

mysql -h db-valentina.czckyeamig9k.us-east-1.rds.amazonaws.com -P 3306 -u valadmin -p'<Enter_DB_Password>' --ssl-verify-server-cert --ssl-ca=/certs/global-bundle.pem mysql

Connected compute resources (0)

Info

Actions

Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.

Filter by compute resources

< 1 >

Resource identifier

Resource type

Availability Zone

VPC security group

Compute resource security group

No connected compute resources

No connected compute resources that were created automatically to display.

Set up EC2 connection

Set up Lambda connection

Security group rules (1)

CloudShell

Feedback

Console Mobile App

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5. Conexión con EC2

```
$ ssh -i "valentina-key.pem" ec2-user@ec2-13-220-17-78.compute-1.amazonaws.com
The authenticity of host 'ec2-13-220-17-78.compute-1.amazonaws.com (13.220.17.78)' can't be established.
ED25519 key fingerprint is: SHA256:j7KcKKXn+K5aXrKcIdnD6izV0cu6ZSiqp+tjjUc4fIA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-220-17-78.compute-1.amazonaws.com' (ED25519)
to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

      #_
     ~\  #####_      Amazon Linux 2023
    ~\  #####\
   ~\  #####|
  ~\  #####/
 ~\  #####/
~\  #####/
 ~\  #####/
  ~\  #####/
   ~\  #####/
    ~\  #####/
     ~\  #####/
      #_

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-4-107 ~]$
```

6. Descarga mariadb

```
[ec2-user@ip-172-31-4-107 ~]$ sudo dnf install mariadb105
Last metadata expiration check: 0:00:14 ago on Fri Feb 27 04:23:49 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing: mariadb105	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	1.5 M
Installing dependencies:				
mariadb-connector-c	x86_64	3.3.10-1.amzn2023.0.1	amazonlinux	211 k
mariadb-connector-c-config	noarch	3.3.10-1.amzn2023.0.1	amazonlinux	9.9 k
mariadb105-common	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	28 k
perl-Sys-Hostname	x86_64	1.23-477.amzn2023.0.7	amazonlinux	16 k

```
Transaction Summary
-----
Install 5 Packages

Total download size: 1.8 M
Installed size: 19 M
Is this ok [y/N]: y
Downloading Packages:
(1/5): mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64.rpm      4.6 MB/s | 211 kB      00:00
(2/5): mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch.rpm 206 kB/s | 9.9 kB      00:00
(3/5): mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64.rpm  1.1 MB/s | 28 kB       00:00
(4/5): perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64.rpm      504 kB/s | 16 kB       00:00
(5/5): mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64.rpm         13 MB/s | 1.5 MB       00:00
-----
Total                                                                    11 MB/s | 1.8 MB       00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Installing     : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 1/5
  Installing     : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      2/5
  Installing     : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64     3/5
  Installing     : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64        4/5
  Installing     : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64           5/5
  Running scriptlet: mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64         5/5
  Verifying      : mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      1/5
  Verifying      : mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch 2/5
  Verifying      : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64           3/5
  Verifying      : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64     4/5
  Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64        5/5

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64          mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64

Complete!
```

7. Acceso base de datos

```
[ec2-user@ip-172-31-4-107 ~]$ mysql -h db-valentina.czckyeamig9k.us-east-1.rds.amazonaws.com -P 3306 -u valadmin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
MySQL [(none)]> CREATE DATABASE modulo2valentina;
Query OK, 1 row affected (0.046 sec)

MySQL [(none)]> SHOW DATABASE
->
-> AC

-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
use near 'DATABASE' at line 1
MySQL [(none)]>
MySQL [(none)]> SHOW DATABASE;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
use near 'DATABASE' at line 1
MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| modulo2valentina |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.008 sec)
```